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Here Come the Suns: Binary Populations in Early Galaxy Formation

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ABSTRACT

With the advent of the James Webb Space Telescope's (JWST) spectroscopy, we can measure the ultraviolet (UV) continuum and nebular emission lines powered by ionizing radiation at redshifts $z \sim 14$. Interpreting these JWST observations at high redshifts requires understanding the spectral energy distribution (SED) of galaxies. Most models used to study structure formations derive teh SED from single, non-interacting stars, neglecting the role binary stars play in star formation history. In close binaries, one star can lose its hydrogen-rich envelope to a companion, leaving behind a hot, stripped helium star. Without their hydrogen envelope, these stripped stars are extremeley hot and emit hydrogen (H) and Helium II (He II) ionizing photons long after the most massive stars have died.

In this work, I will add stripped bianry populations to a semi-analytical model called `DELPHI`, reconstruct the star-formation history of every `DELPHI` galaxy with a stripped binary population, and study the effects of stripped binaries on the SED of early galaxies. From the resulting SED, I will quantify the effects of stripped binaries on the UV luminosity function (UV LF), the UV luminosity density (UV LD), the UV slope (β), the production rate of H and He II ionization photons, and the ionization efficiency ξ_{ion}.

1. INTRODUCTION

The James Webb Space Telescope (JWST) has revolutionized the field of high-redshift galaxy formation by offering groundbreaking sensitivity, spatial resolution, and wavelength coverage. It has enabled astronomers to identify hundreds of star-forming galaxies at redshifts beyond $z \sim 14$, rattling the scientific community with two interesting observations: an unexpected abundance of super-early, massive galaxies at the bright end ($M_{UV} \approx -21$) of the ultraviolet luminosity function (UV LF); and a growing number of sources that show hard ionizing spectra with strong He II emission. Both observations constrain the intrinsic stellar SED of early galaxies.

To model and predict early galaxy formation, one must model the growth of matter in early Universe. Fortunately, the cosmological principle suggests that on large scales, the early Universe is homogeneous and isotropic, and the first bound structures are low-mass dark matter halos that collapse into overdense regions. Due to their shallow potential wells and low masses, even a small amount of star formation can provide the necessary feedback to expel all star-forming gas from these systems. To resume star formation, a low-mass halo must, then, wait for gas to be accreted from the surrounding intergalactic medium (IGM) or to be deposited via mergers. Dayal et al. (2014) developed a semi-analytical model called `DELPHI` that embeds this simple principle: at any given time, there exists a limiting star formation efficiency such that the energy produced by a newly formed star is sufficient to expel all remaining gas, at least temporarily.

In most models like `DELPHI` , it is assumed that stellar populations consist of single stars. At each time step, single-star population-synthesis tables are used to evaluate the light of a stellar population at a given age and metallicity. Therefore, at a chosen initial mass function, that table fixes the 1500 Å luminosity and the production rate of hydrogen-ionizing ($E_\gamma > 13.6$ eV) and He-II ionizing ($E_\gamma > 54.4$ eV photons. However, binary interactions via mass transfer open a new possibility.

For instance, a stripped star is a hot, helium-rich core left after a binary companion removed the hydrogen envelope. These stars emit many photons with $E_\gamma >$ 13.6 eV (hydrogen-ionizing) and even $E_\gamma > 54.4$ eV (helium-ionizing). Since stripped stars are so hot, they should also be able to add some light at 1500$\mathring{A}$. Their contributions are dependent on metallicity. At very low metallicity ($Z \sim 0.0002$, there is a thin hydrogen layer on the star, making it cooler and reducing the amount of He II photons that are released. This work aims to add the stripped binary population to DELPHI and determine how it changes the SED of stellar populations during early galaxy formations.

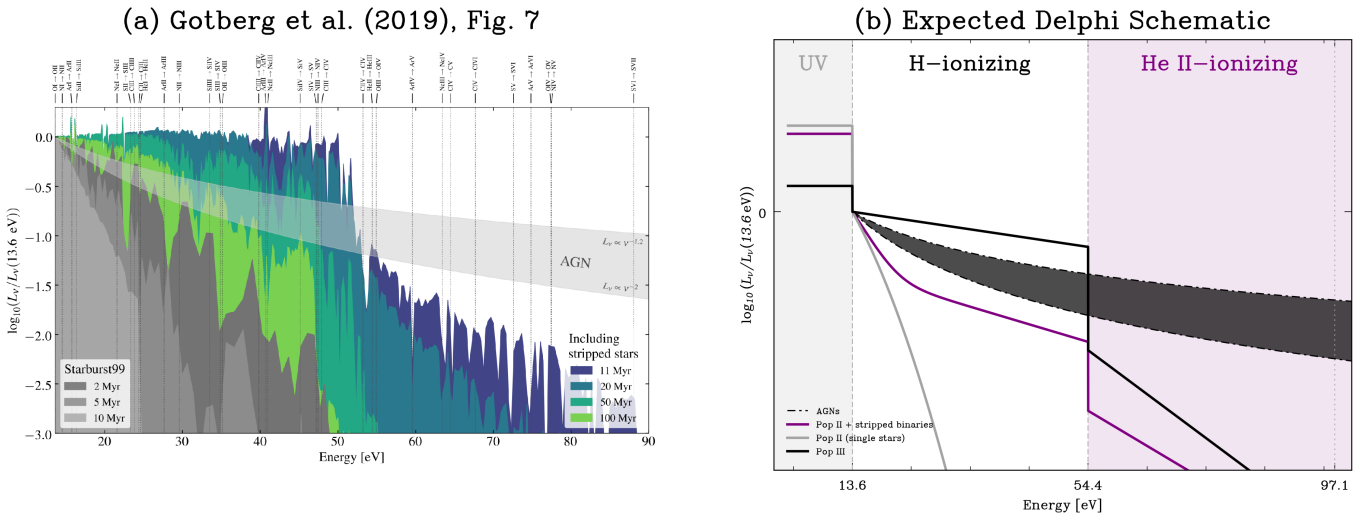


Figure 1. *Left/Panel (a)* This is Figure 7 of Götberg et al. (2019). We see the ionizing part of the spectra of co-eval stellar populations at solar metallicity, for single stars only (Starburst 99, gray), and including stripped stars (at different ages from burst in colors) with AGN slopes $L_\nu \propto \nu^{-1.2} - \nu^{-2}$ in the gray band. *Right/Panel(b)* This is the schematic of the expected SED from this work. The curves in this panel are purely illustrative and not indicative of actual, scientific results. Pop II single stars are in gray, Pop II with stripped binaries are in purple, Pop III stars are in black and the AGN population is the black band.

2. SED OF AN EARLY GALAXY

We can divide the SED into three separate regions for the purposes of this work

- **UV - non-ionizing**. $E_\gamma < 13.6\,eV$ and it can be seen directly through the UV luminosity function, the UV luminosity density and the UV slope β.
- **H-ionizing** $E_\gamma > 13.6\,eV$ and these photons are absorbed by the gas and seen through hydrogen recombination lines (Lyα, Hα, Hβ). We can use this region to find the photon rate $Q(H)$ and $\xi_{ion} = Q(H)/L_{UV}$.
- **He II-ionizing** $E_\gamma > 54.4\,eV$ and these photons ionize He^+. They are seen through He II λ1640 and λ4686. We can use this region to find the photon rate Q(He II) and the hardness $Q(HeII)/Q(H)$.

Stripped stars should add only a very small fraction to the UV window, while massive stars are still alive. It is once the massive stars cease to exist that stripped stars continue to emit ionizing photons; the ionizing emission of stripped stars dominates the ionizing output of populations a few MYrs after star formation ceases.

A schematic of the anticipated SED, inspired by Figure 7 in Götberg et al. (2019), is presented in Figure 1. In panel (a) of Figure 1, we reproduce Figure 7 from (Götberg et al. 2019), which already shows the stripped stars hardening the ionizing spectrum of a stellar population. Panel (a) shows the ionizing spectrum for a population formed in a single instantaneous burst (co-eval stellar population). A real galaxy is neither a single burst or a population with a constant star formation rate. In `DELPHI`, each galaxy is the sum of multiple bursts and its SED is the weighted sum of SEDs similar to panel (a). Panel (b) is the expected schematic we want to achieve after incorporating stripped binaries into realistic star formation histories.

2.1. *Star Formation History to a SED*

The stellar SED of a galaxy in `DELPHI` is built from its star-formation history in three steps (Figure 2):

- `DELPHI` stores, in 30 Myr steps, the stellar mass formed in each halo $M_*^{new} = SFR \times \Delta t$, and the metallicity of the gas it formed from.
- At each burst, we can find the SED, per unit SFR, of stars formed at a constant rate during one step and at that age. We find the luminosity per unit star formation rate for single stars (from Starburst99) and stripped stars: l_λ^{single} and l_λ^{strip} respectively.
- Sum over all bursts k, where burst k formed n_k steps before galaxy j is observed. The sum is the total SED given by

$$L_\lambda(j) = \sum_k SFR_k [l_\lambda^{single}(n_k, Z_k) + l_\lambda^{strip}(n_k, Z_k)] \quad (1)$$

The first stripped stars appear around 10 Myr after a burst (Götberg et al. 2019). As panel (b) in Figure 2 goes up to 890 Myr in `DELPHI` , stripped stars contribute ionizing photons for a while. In panel (c) of Figure 2, we

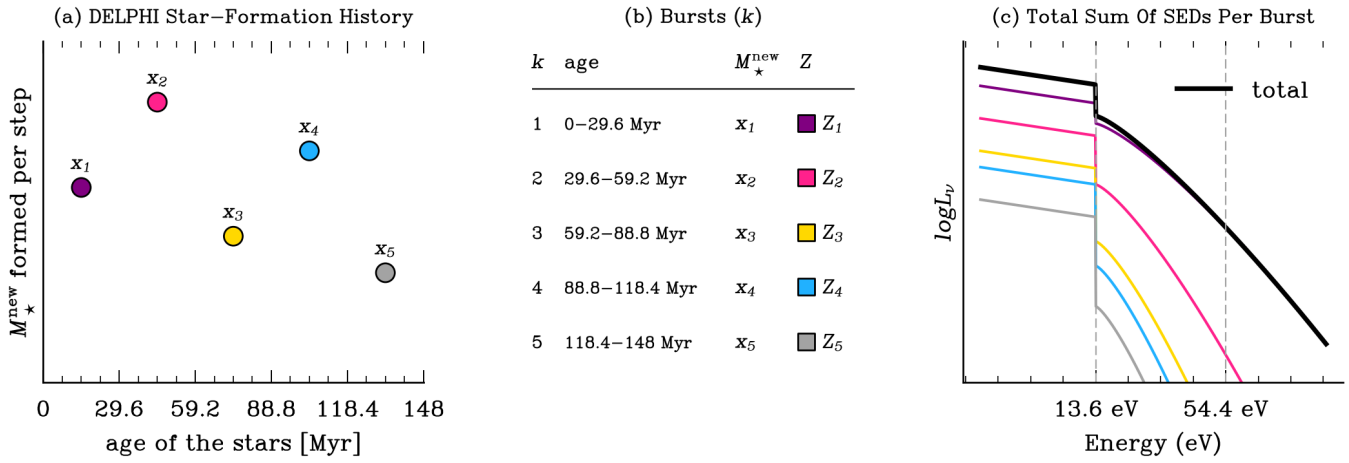


Figure 2. We show how SEDs are built in `DELPHI` through this illustration. *Left/Panel (a)* The stellar mass formed in each 30 Myr step inside `DELPHI` are labeled as $x_1, x_2, ..., x_5$. The colors represent the metallicity of the gas. *Center/Panel (b)* This shows different rows for different bursts (k). *Right/ Panel (c)* This shows the SED of each burst, summed up to give the total SED of the galaxy in black.

also see the SEDs of the older bursts (like $x_2 - x_5$) drop sharply above $13.6eV$ once the massive stars have died. Adding stripped binaries will give the SED of the older bursts a trail.

Furthermore, once we find the SED, we can investigate Q(H), Q(He II) and ξ_{ion} to see how they change.

2.2. *Where do He-II ionizing photons come from?*

In single-star Pop II models, photons above 54.4 eV are rare, so nebular He II in a galaxy with a Pop II population is usually attributed to sources like AGN, accreting black holes, X-ray binaries (Schaerer et al. 2019), or, at high redshift, Pop III stars. Even in metal-poor star-forming galaxies, He II is often stronger than single-star population models predict (Shirazi and Brinchmann 2012; Senchyna et al. 2017). Stripped binaries are missing from single-star models, yet they emit He II-ionizing photons (panel (a) in Figure 1. An outcome of this work is to answer whether some of the He II that is currently thought to originate from AGN, black holes or Pop III stars could actually be from stripped stars.

3. IMPACT OF STRIPPED BINARIES ON SEARCH FOR THE FIRST STARS

Hard ionizing radiation, traced by nebular He II emission, is the best signature of Pop III stars (Venditti et al. 2026). Figure 3 shows why this signature is difficult to use in practice. A Pop III-hosting galaxy passes through a pristine phase, a self-polluted phase (where it is pure Pop III), and a hybrid phase in which a Pop II population forms alongside the surviving Pop III stars. The Pop III-dominated phase($M_{\star,PopIII} > 0.5M_{\star,tot}$) lasts only $\sim$ 15-20 Myr.

In Figure 3, on the EW(HeIIλ4686) versus [OIII]λ5007/$H\beta$ diagram in Figure 3, the self-polluted and Pop III-rich galaxies lie above the Pop III threshold established in Nakajima and Maiolino (2022). Pop III-mid galaxies spread across the threshold, and Pop III-poor galaxies fall well below it, close to the region where AGN and other hard sources produce similar line ratios. Stripped helium stars are among these contaminants (Venditti et al. 2026), and they are one that `DELPHI` can model.

Stripped stars in the Pop II component of a hybrid galaxy can raise EW(He II) and so create false Pop III candidates. In a hybrid galaxy, the Pop II component contains binaries whose stripped stars keep emitting He II ionizing photons after its massive stars have died (Götberg et al. 2019). This can raise EW(He II) without any Pop III stars and push Pop III-poor or Pop II-only galaxies towards the Pop III threshold, creating false Pop III candidates.

This contamination should be strongest in enriched gas, and should depend strongly on metallicity since the temperature of the stripped stars falls at low metallicity. At Z$\sim$0.0002, binary stripping leaves a thin hydrogen layer on thestar, making it cooler, so its He II ionizing output is about 100 times lower than at higher metallicity (Götberg et al. 2019). In addition, massive low-metallicity stars do not expand enough to be stripped, so the most massive and hottest stripped stars do not form (Hovis-Afflerbach et al. 2025) Modeling stripped stars in Delphi yields the He II emission that Pop II populations alone can produce as a function of metallicity and redshift. This new boundary or floor will help determine whether a He II detection is appropriate to strengthen

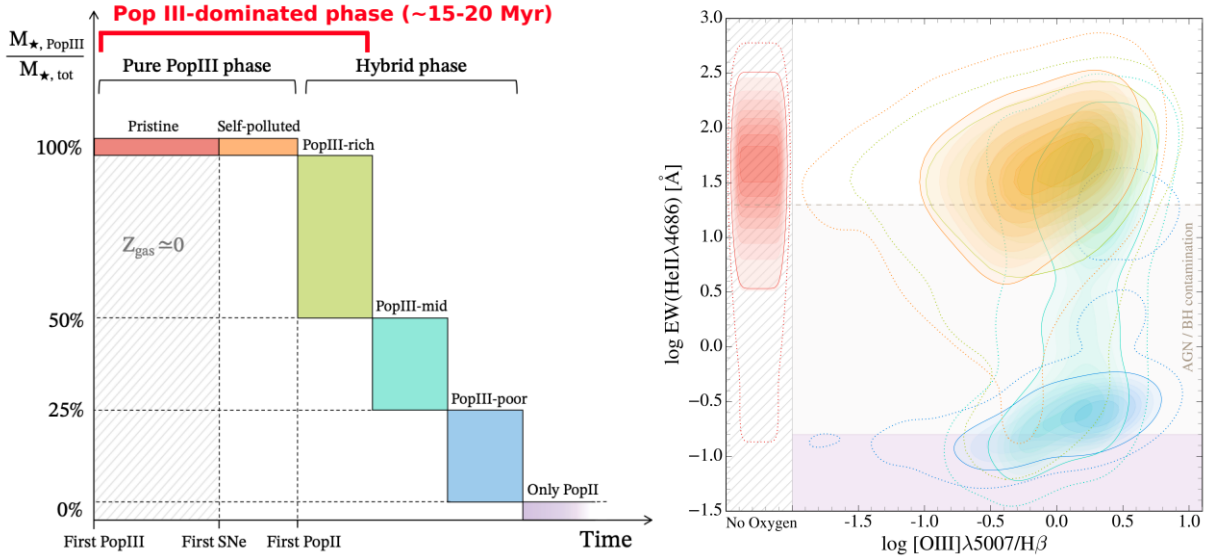


Figure 3. Fig. 10 of Venditti et al. (2026), adapted from Rusta et al. (2025). *Left:* evolutionary stages of a Pop III-hosting galaxy, ordered by the Pop III fraction of its stellar mass. The Pop III-dominated phase ($M_{\star,PopIII} > 0.5\,M_{\star,tot}$) lasts ∼15-20 Myr. *Right:* Distribution of Pop III-hosting galaxies predicted by the `NEFERTITI` model on the EW(He IIλ4686) versus [OIII] λ5007/Hβ diagram. Colors match the stages on the left; solid (dotted) contours enclose 68% (95%) of the galaxies. The dashed line marks the Pop III threshold of **?**, and the lilac region below it the zone contaminated by AGN and direct-collapse black holes.

the case for a Pop III star or not. A He II detection above the floor is a Pop III star, while a detection below it is a Pop II contaminant masquerading as Pop III.

4. MILESTONES AND TIMELINE

I propose the following between the months of October to May for my AST1501Y project, supervised by Prof. Pratika Dayal. In my timeline, I have taken into account the time required to learn `DELPHI`, meet 1501Y deadlines and conduct actual research. A gantt chart to summarize this timeline is displayed in Figure 4. Note that a buffer time for minor existential crises expected around the holidays were not included in the timeline.

- **October** - I have already learned Delphi, compiled it using five seeds for the fiducial, eSFE and eIMF models on the 600 trees. I have reproduced the UV luminosity function, UV luminosity density, halo mass function and stellar mass function plots from Mauerhofer et al. (2025), and reproduced them in this proposal through Figures **??**, **??**, **??** and **??**.

- **November** - I will add the stripped-star UV grid to calculate the UV results. I will also be working on the peer feedback, proposal, and presentation preparation during this period.

- **December** - I will finish the integration of stripped stars and their UV spectra from Götberg et al. (2019) into `DELPHI`. I will then download the catalog from Hovis-Afflerbach et al. (2025) to find the fraction of stripped-binary against metallicity.

- **January** - I will start reviewing line strengths, finish the stripped-binary fraction plot against metallicity, decide which lines to model (e.g. $H\alpha, H\beta$ and how to treat escape and ionization parameters. I will also prepare the 1501Y midpoint update.

- **February** - Finish the line strengths for each halo with and without stripped stars. Turn them into observable predictions in JWST (equivalent widths, line ratios, line luminosity functions) at the redshifts used for the UV results.

- **March** - I will concentrate on finishing the JWST ;line predictions this month and start writing the final 1501 report.

- **April** - I will prepare and give the 1501 final presentation, evaluating the results of my work and finding the floor that separates fake Pop III stars from the real ones.

5. ACKNOWLEDGMENTS

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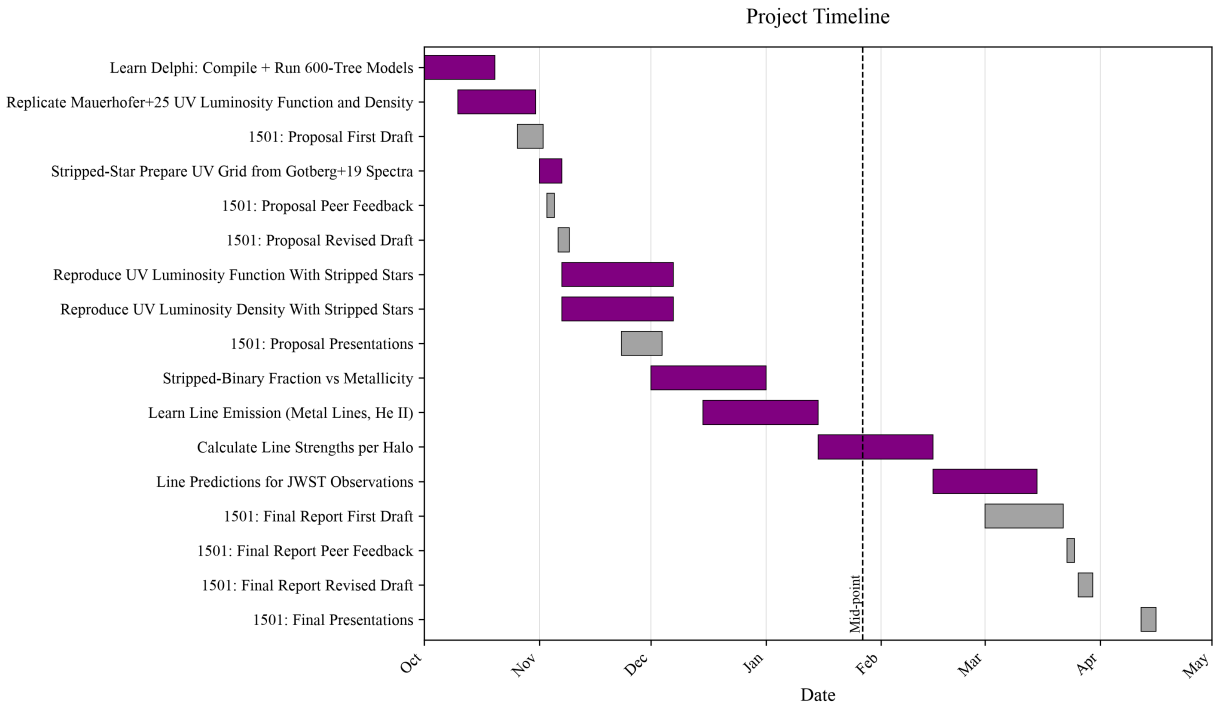


Figure 4. Gantt Chart

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